

VISHAL KOTHA

MATERIALS CHEMISTRY · ELECTROCATALYSIS · GREEN HYDROGEN · NANOFABRICATION
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11 peer-reviewed articles · 189 citations · h-index 8 · i10-index 7 · 2 as corresponding author

RESEARCH FOCUS

Defect-engineered perovskite electrocatalysts for industrial green-hydrogen production — bridging atomic-scale materials design with real-world electrolyzer deployment.

Keywords: Perovskite oxides · OER/HER electrocatalysis · Green hydrogen · Zinc-air batteries · CISS / spin-selective catalysis · Electron microscopy · Nanofabrication

PROFESSIONAL SUMMARY

Materials chemist (Ph.D., IIT Bombay) with 7+ years across academia and industry in nanomaterials, thin films, and electrocatalysis for clean energy. Currently Scientist-B at BPCL Corporate R&D, leading a commercial-scale alkaline water electrolyzer program for green hydrogen (OER/HER and AOR/HER systems). Author of 11 peer-reviewed papers (189 citations, h-index 8), including J. Mater. Chem. A and ACS Appl. Energy Mater., with deep hands-on mastery of advanced characterization (TEM, XPS, XRD/Rietveld, electrochemistry) and nanofabrication (e-beam/photolithography, ALD, CVD). An international, intersectoral trajectory — IIT Bombay to the Weizmann Institute to industrial R&D — positioning discovery for real-world scale-up.

EDUCATION

Ph.D. in Chemistry, Indian Institute of Technology Bombay · 2016–2022 · CPI 8.44/10

Thesis: Tailoring Transition Metal Perovskite Oxides via Low-Temperature Hydrothermal Routes as Potential Candidates for Catalytic Applications. Advisor: Prof. Leela Srinivas Panchakarla (late).

M.Sc. in Chemical Sciences, Pondicherry University · 2014–2016 · CGPA 8.4/10

Thesis: On the Origin of Chink in B₄C Armor — Anatomy of a classical 2c-2e B–B bond. Advisor: Prof. Musiri M. Balakrishna Rajan.

B.Sc. in Chemistry (Hons.), Khallikote University, Odisha · 2011–2014 · 85.4% · **Gold Medalist**

PROFESSIONAL EXPERIENCE

2026–Present



Scientist-B (FTE) Bharat Petroleum Corporation Ltd. · Greater Noida, India

- Leading the **alkaline water electrolyzer** program for green hydrogen — commercial-scale OER/HER and AOR/HER electrocatalyst systems for industrial water electrolysis.
- Contributing to an **electrochemical H₂S valorization** project — sulfide-based anodic reactions as a lower-energy alternative to conventional OER.
- Developing **retardant catalyst materials for a ceramic hydrogen burner** for process-safety and combustion-control research.

2024–2025



Science Educator (Residential Program) Isha Foundation · Coimbatore, India

- Delivered structured **ICSE Chemistry (Grades 9 & 11)** and secondary Science & Mathematics within an immersive residential program.
- Strengthened classroom pedagogy, curriculum pacing, and one-on-one mentoring — foundational to a professorial teaching practice.

2023–2024



Postdoctoral Fellow Weizmann Institute of Science · Rehovot, Israel

- Pioneered **chirality induction in Ni–Au systems** for the hydrogen evolution reaction (HER), improving efficiency ~10% for green-fuel applications; characterized by TEM and XPS.
- Collaborated with international teams to demonstrate superior HER activity via spin-selective (CISS) catalysis.

2022–2023



Postdoctoral Fellow Indian Institute of Technology Bombay · Mumbai, India

- Investigated magneto-electric effects on water electrolysis using **La₂FeMnO₆**, advancing multiferroic catalysts for energy conversion.
- Developed mechanochemical shear-induced nucleation and exfoliation of nanomaterials for

2016–2022



scalable synthesis.

Ph.D. Researcher Indian Institute of Technology Bombay · Mumbai, India

- Led hydrothermal synthesis of **K-substituted LaMnO₃** perovskite for zinc–air batteries and organocatalysis — 3 high-impact publications.
- Operated and trained peers on FEG-TEM, SEM-EDS, PXRD, ICP-AES (**500+ hours**); mentored 4 students.
- Fabricated MOSFET devices from 1D nanorods, exploring interconnects for energy-efficient electronics.

PUBLICATIONS

*V. Kotha shown in bold. * denotes the two publications where V. Kotha is corresponding author.*

1. Nishad, H. S.; **Kotha, V.**; Sarawade, P.; et al. Exchanging interlayer anions in NiFe-LDH nanospheres enables superior battery-type storage for high-rate aqueous hybrid supercapacitors. *J. Mater. Chem. A* 2024, 12 (16), 9494–9507.
2. Gupta, S. P.; **Kotha, V.**; Walke, P. S.; Panchakarla, L. S. Hydrated WO₃·0.33H₂O nanorod anodes for high-performance, long-cycle aqueous Zn-ion batteries. *J. Power Sources* 2024, 604, 234500.
3. Nishad, H. S.; Gupta, S. P.; **Kotha, V.**; et al. Sodium-substituted tungsten oxide nanoflowers for enhanced pseudocapacitive storage in aqueous asymmetric supercapacitors. *ChemNanoMat* 2024, 10 (2), e202300463.
4. **Kotha, V.***; Karajagi, I.; Ghosh, P. C.; Panchakarla, L. S. Potassium-substituted LaMnO₃ as a highly active, exceptionally stable bifunctional ORR/OER electrocatalyst. *ACS Appl. Energy Mater.* 2022, 5 (6), 7297–7307.
5. **Kotha, V.***; Reddy, P. L.; Maddu, D.; Panchakarla, L. S. Importance of K substitution in LaMnO₃ nanostructures for selective reduction of α,β -unsaturated carbonyls and nitroarenes. *ACS Appl. Nano Mater.* 2022, 5 (12), 17482–17486.
6. **Kotha, V.**; Kumar, K.; Dayman, P.; Panchakarla, L. S. Doping with chemically hard elements to improve photocatalytic properties of ZnO nanostructures. *J. Clust. Sci.* 2022, 33 (5), 1943–1950.
7. Bhavani, B.; Chanda, N.; **Kotha, V.**; et al. 1D alignment of Co(II)-metalated porphyrin–naphthalimide self-assembled nanowires for photocatalytic hydrogen evolution. *Nanoscale* 2022, 14 (1), 140–146.
8. Nishad, H. S.; Gupta, S. P.; **Kotha, V.**; et al. Enhanced van der Waals separation in hydrated tungsten oxide nanoplates for superior pseudocapacitive charge storage. *J. Alloys Compd.* 2022, 914, 165227.
9. Bera, A.; Pathak, S. S.; **Kotha, V.**; Prasad, B. L. V. Lamellar bimetallic thiolates for mechanochemical preparation of bimetallic chalcogenide nanocrystals. *Adv. Mater. Interfaces* 2021, 8 (23), 2100898.
10. Sugathan, V.; Ghosh, B.; Harikesh, P. C.; **Kotha, V.**; et al. Synthesis of bismuth sulphoiodide thin films from a single-precursor solution. *Sol. Energy* 2021, 230, 714–720.
11. Mrinalini, M.; Suman Krishna, J. V.; Krishna, N. V.; **Kotha, V.**; et al. Photobleaching of triphenylamine-phthalocyanine via mixed valence-state-triggered self-assembled nanospheres. *J. Phys. Chem. C* 2018, 122 (34), 19946–19952.

TECHNICAL SKILLS

- **Synthesis:** Hydrothermal, mechanochemical, solid-state, solution-based, electroplating, CVD.
- **Characterization:** TEM (HR, SAED, EDAX), SEM-EDS, XRD with Rietveld refinement, XPS, AFM, ICP-AES; CV, LSV, impedance, ellipsometry.
- **Nanofabrication:** Electron-beam lithography, photolithography, sputtering, ALD.
- **Computation & Software:** Python (data analysis), VASP, Gaussian, FullProf, PANalytical HighScore+, OriginLab, ChemDraw, ImageJ.

TEACHING EXPERIENCE

- Teaching Assistant, **Electrochemistry (CH117L)**, Dept. of Chemistry, IIT Bombay (2016–2018) — 30/30 lab sessions.
- ICSE Chemistry (Grades 9 & 11), Isha Home School, Coimbatore (2024).
- Science & Mathematics (secondary), Isha Samskriti, Coimbatore (2024–2025).
- Visiting Tutor in Chemistry, Dawn Breeze International School, Berhampur (2012–2013).

MENTORSHIP & INVITED LECTURES

- **Mentored:** INSPIRE Summer Fellow (2019); M.Sc. theses — Pawan Dayman (2018–19), Bhaskar (2020); B.Sc. thesis — Divya Maddu (2019–20).
- **Invited talk:** “Future of Chemistry: Exploration of the Possibilities,” Khallikote University (Mar 2023).
- **Invited talk:** “Insights into the Research Journey — from PhD to post-PhD,” Khallikote University (Feb 2024).

LEADERSHIP & SERVICE

- **Instrument facility management, IIT Bombay (2017-2023):** FEG-TEM (500+ hours), PXRD/Rietveld (20+ samples), FESEM-EDS — sample prep, imaging, and elemental analysis.
- Trained 3 Ph.D. students in TEM/SEM operation; formally mentored 1 Ph.D., 2 M.Sc., 1 B.Sc. researcher.
- Student governance: Mess Secretary and Hostel Hygiene Committee, IIT Bombay (2018).

AWARDS, FELLOWSHIPS & CONFERENCES

- **GATE (Chemistry) 2016:** 97.06 percentile, AIR 440 — Ministry of Education, Govt. of India.
- **JAM 2014:** AIR 1080 — Joint Admission Test for M.Sc.
- **Gold Medalist,** B.Sc. (Hons.) Chemistry, for academic excellence.
- Posters: ICCFM-2018 (S.N. Bose Centre, Kolkata); 15th Indian Theoretical Chemistry Symposium (Univ. of Hyderabad, 2016). Participant: Graphene Symposium & Bengaluru-India Nano Roadshow, IIT Bombay.

REFERENCES

- **Prof. Bhagavatula L. V. Prasad** — Director, CeNS, Bengaluru · pl.bhagavatula@ncl.res.in
- **Prof. Anindya Datta** — Professor, IIT Bombay · anindya@chem.iitb.ac.in
- **Dr. Seelam Prasanth Kumar** — Scientist, CSIR-IICT · prasanth@iict.res.in